Gas quadrangle — Solution

X20 (2020) — English translation

A1 **0.80 pts**

Let the pipe be adiabatically insulated. Obtain the dependence of gas velocity on its temperature.

Answer: \textbfAnswer: ZSE

$$P_0 dV_0 - PdV = dm\left(\frac{v^2 - v_0^2}{2} + \frac{C_V}{\mu}(T - T_0)\right)$$

A2 **0.70 pts**

Show that when gas flows, the relation

$$v\frac{dv}{dx} = -\frac{1}{\rho}\frac{dP}{dx}.$$

is satisfied

The law of momentum change for gas

$$-Sdp = adm$$

Equality $a = v\frac{dv}{dx}$ proves the required

A3 **0.50 pts**

Now thermal power $N = kL$ is supplied to any section of pipe with length L. Derive an expression for the gas velocity as a function of x and T.

ZSE

$$Ndt = P_0 dV_0 - PdV = dm(\frac{v^2 - v_0^2}{2} + C_V(T - T_0))$$

After transformations,we get

$$v^2 = v_0^2 + \frac{2C_P}{\mu}(T_0 - T) + \frac{2N}{m}$$

Answer

$$v = \sqrt{v_0^2 + \frac{2C_P}{\mu}(T_0 - T) + \frac{2kx}{\rho_0 S_0 v_0}}$$

A4 **0.50 pts**

Now the gas temperature depends linearly on x according to the law $T = T_0 + \alpha x$. Show that the gas moves through the pipe with uniform acceleration and find this acceleration a.

$a = v\frac{dv}{dx} = -\frac{\alpha C_P}{\mu} + \frac{k}{\rho_0 S_0 v_0} = const$, which proves the required